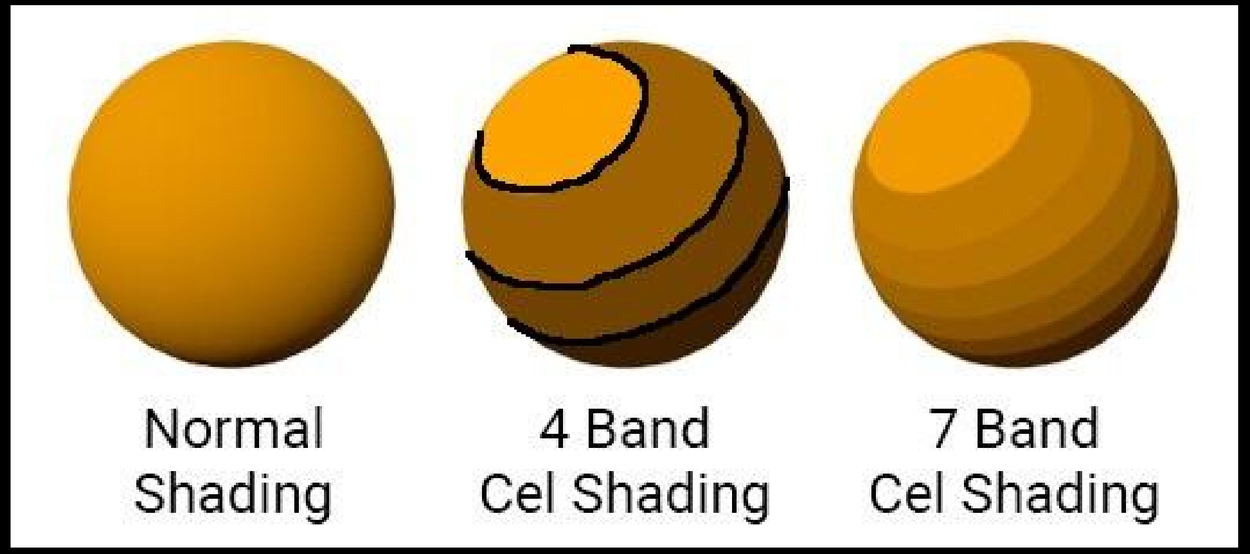
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Procedural Generation Techniques

PCG 03: Shaders

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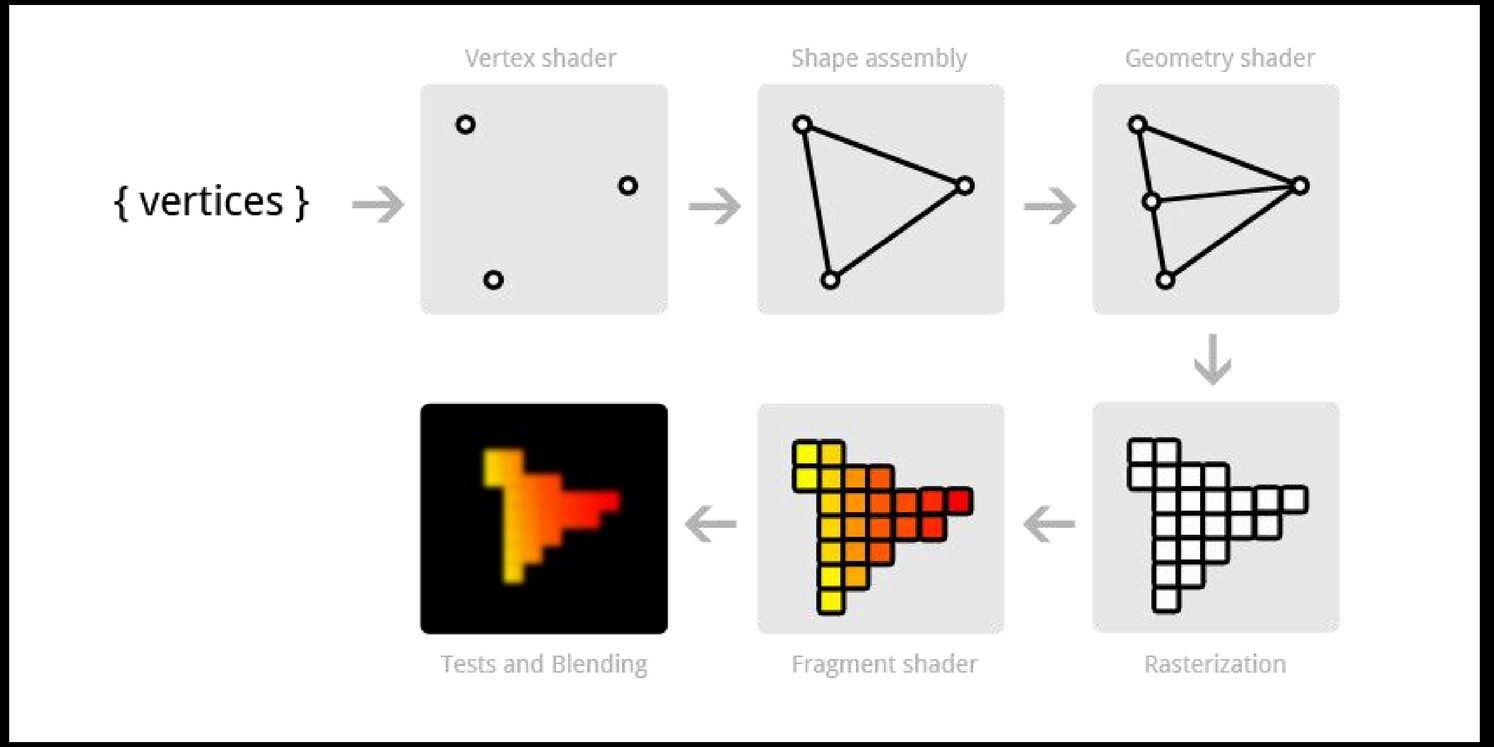
What are shaders:



A shader is a small program that runs on the GPU, once per vertex or once per pixel, in parallel, with no memory of other vertices or pixels. You write the rule for one element; the GPU runs it millions of times at once.

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What are shaders:

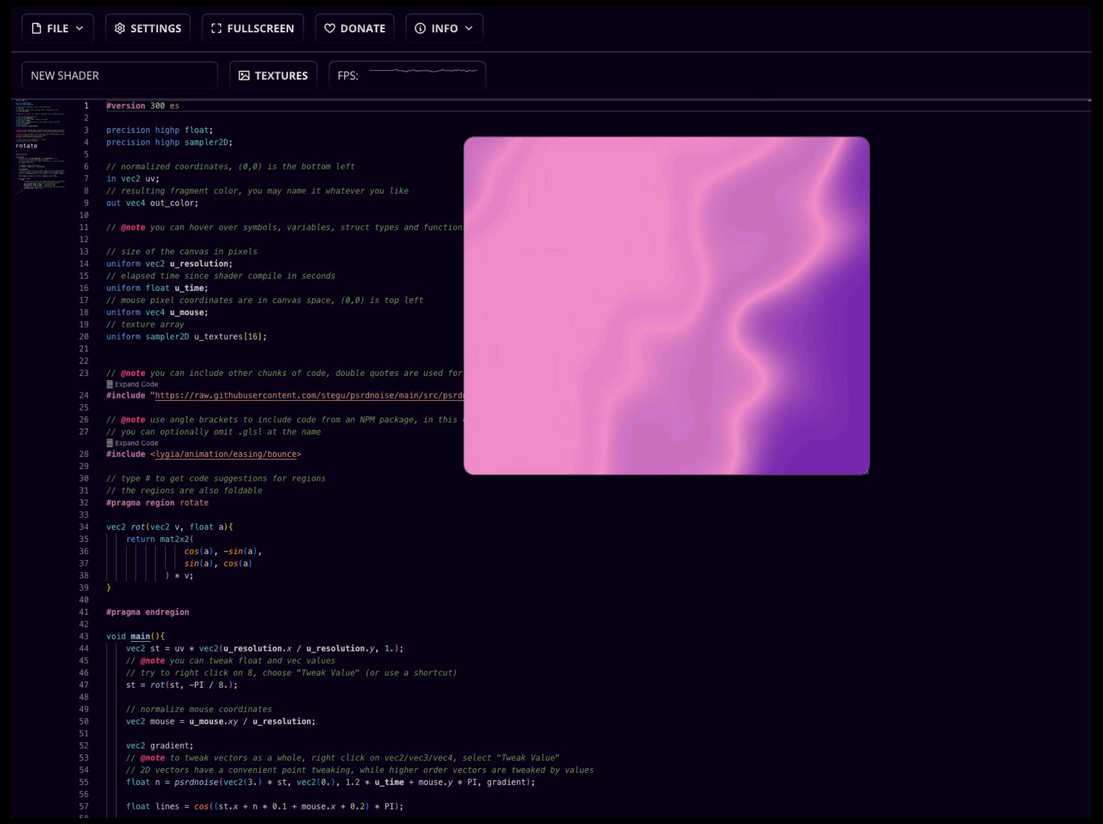


Two parts to a shader:

- **Vertex shader** — runs once per vertex. Its required job is to say where the vertex lands on screen. It can also move vertices (displacement, wobble) and pass values on to the fragment stage.
- **Fragment shader** — runs once per pixel-sized fragment covered by a triangle. Its job is to output a colour.

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Shader language:



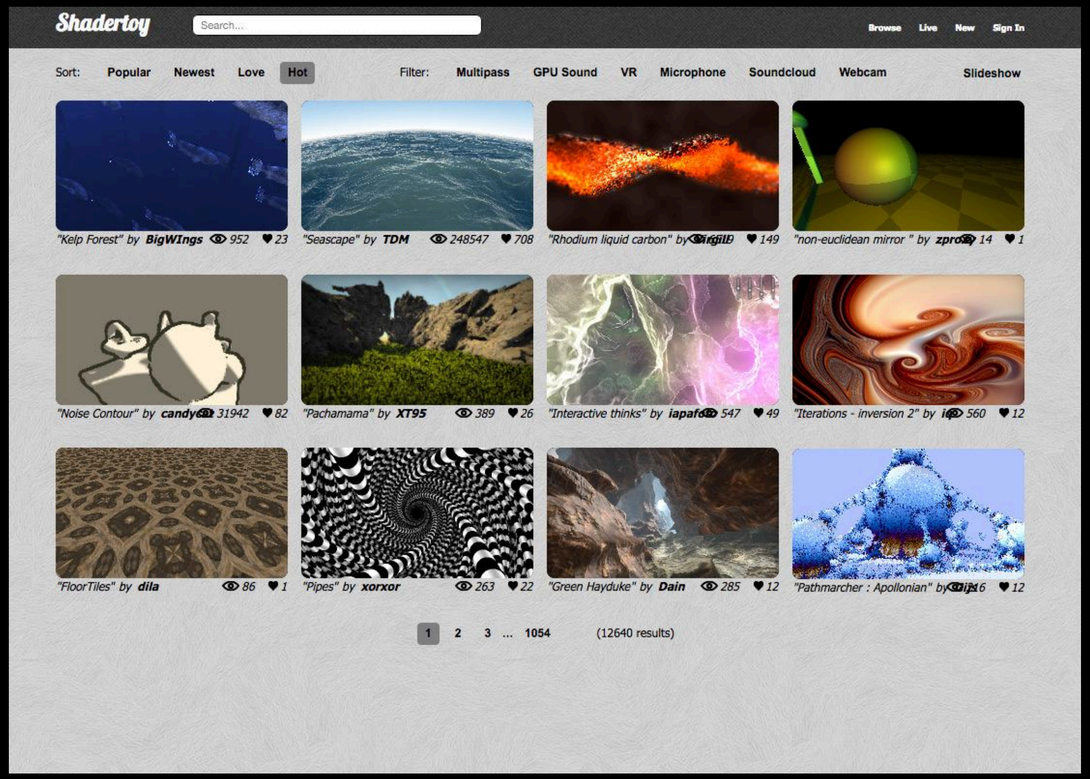
The main languages:

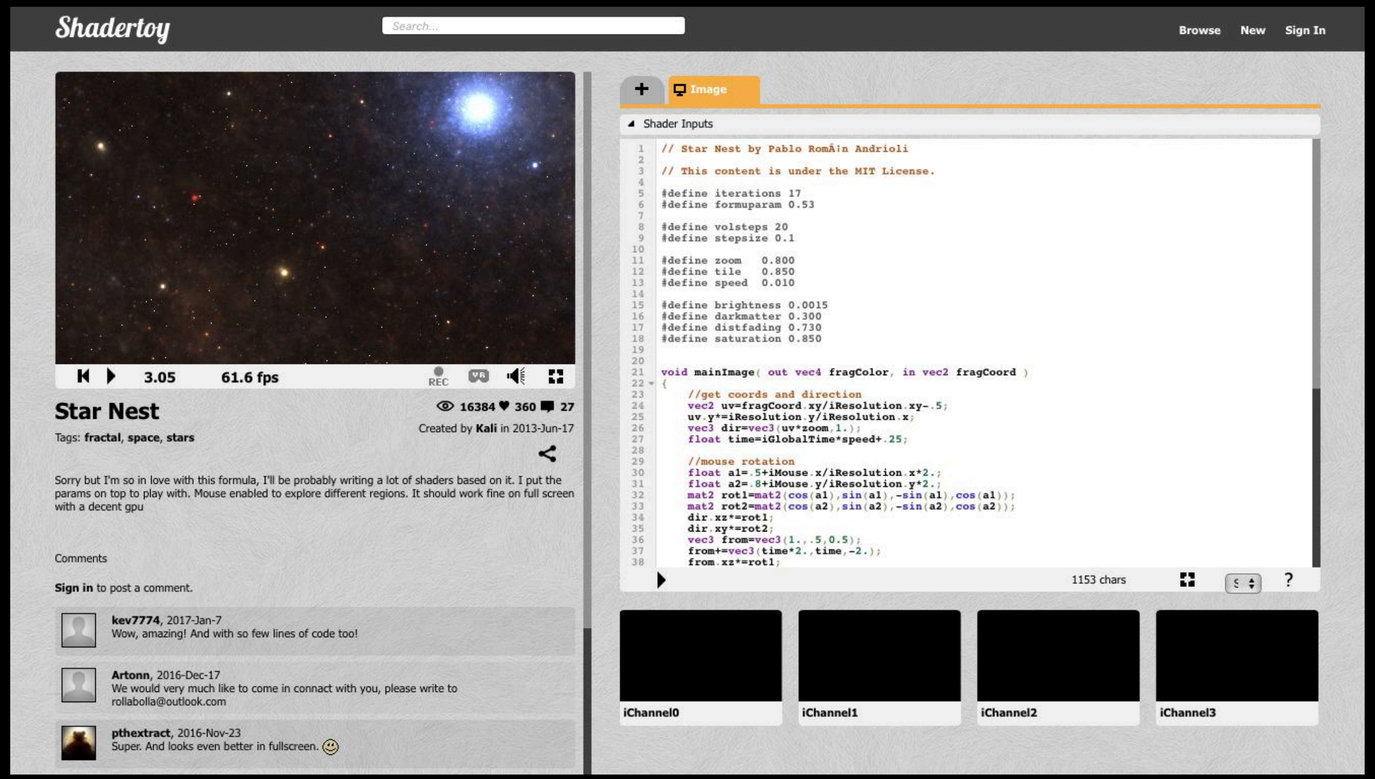
GLSL — OpenGL / WebGL / Vulkan. The lingua franca of tutorials and Shadertoy.

HLSL — DirectX, Unity, Unreal. MSL — Metal, on Apple platforms.

WGSL — WebGPU, the web's new standard. SPIR-V — not written by hand: the compiled binary format the others target. Slang — newer open language that compiles to all of the above.

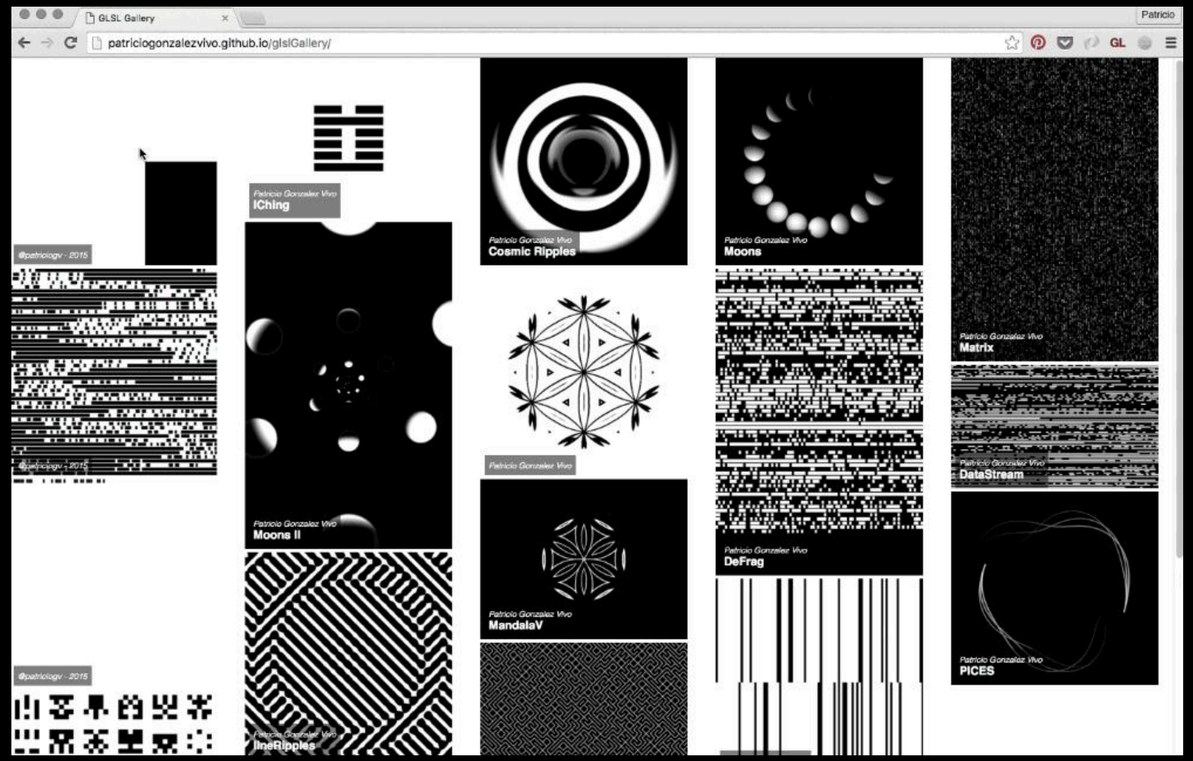
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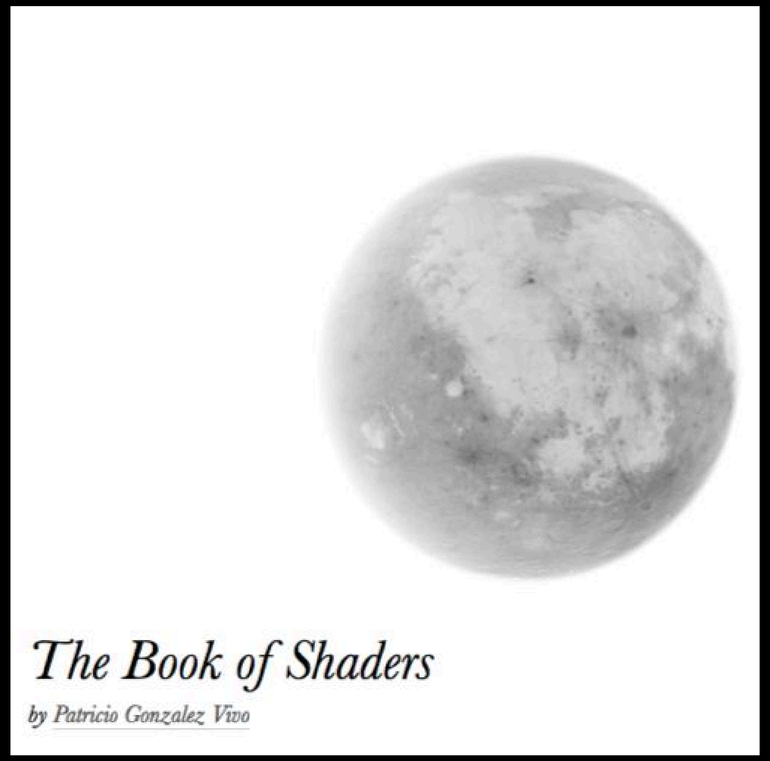




Environments to design shaders:
-Shader Toy
-Shader Graph

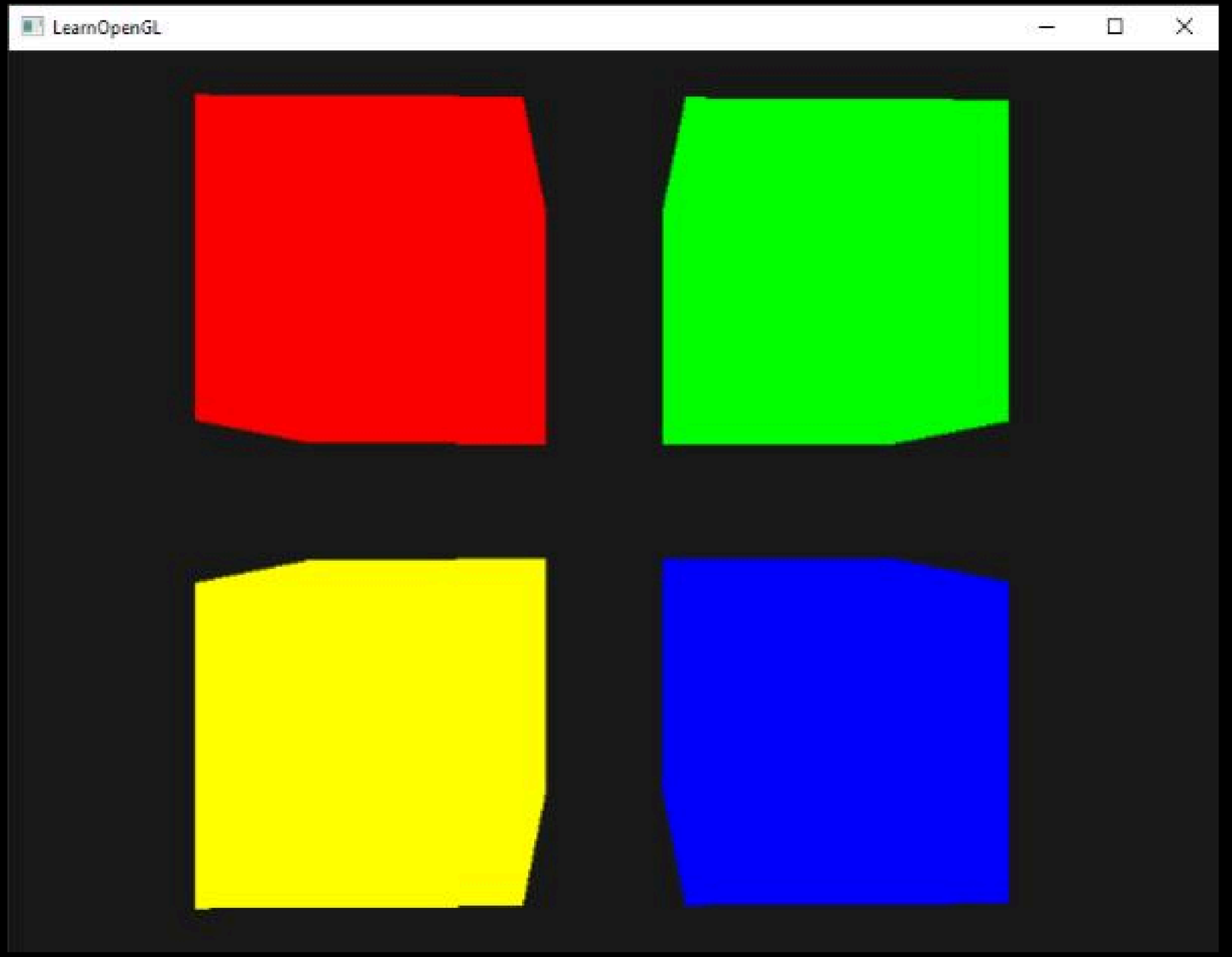
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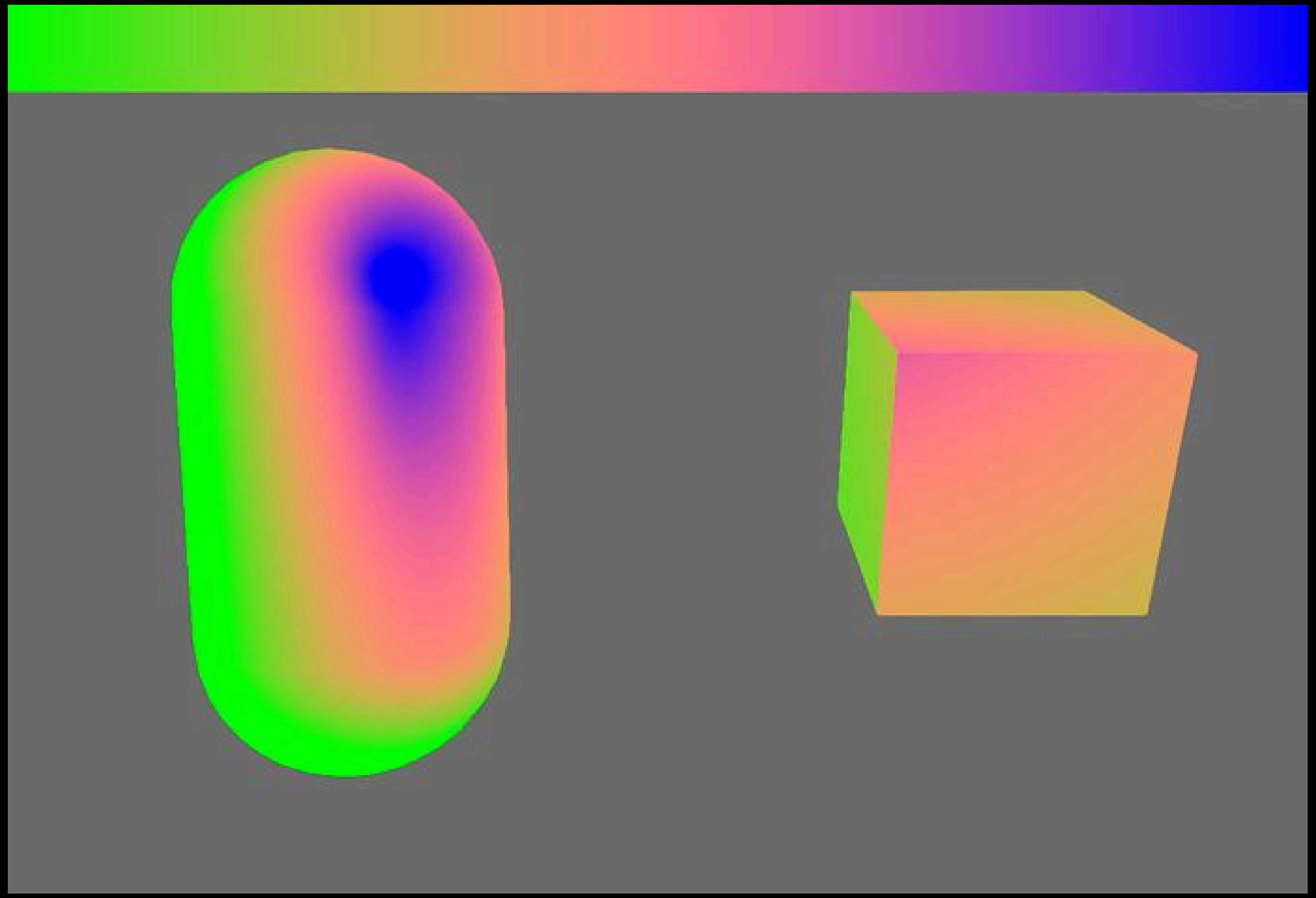
Environments to design shaders:
-The Book of shaders - Patricio Gonzalez Vivo

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Shader examples:
Flat Shader (Uniform)

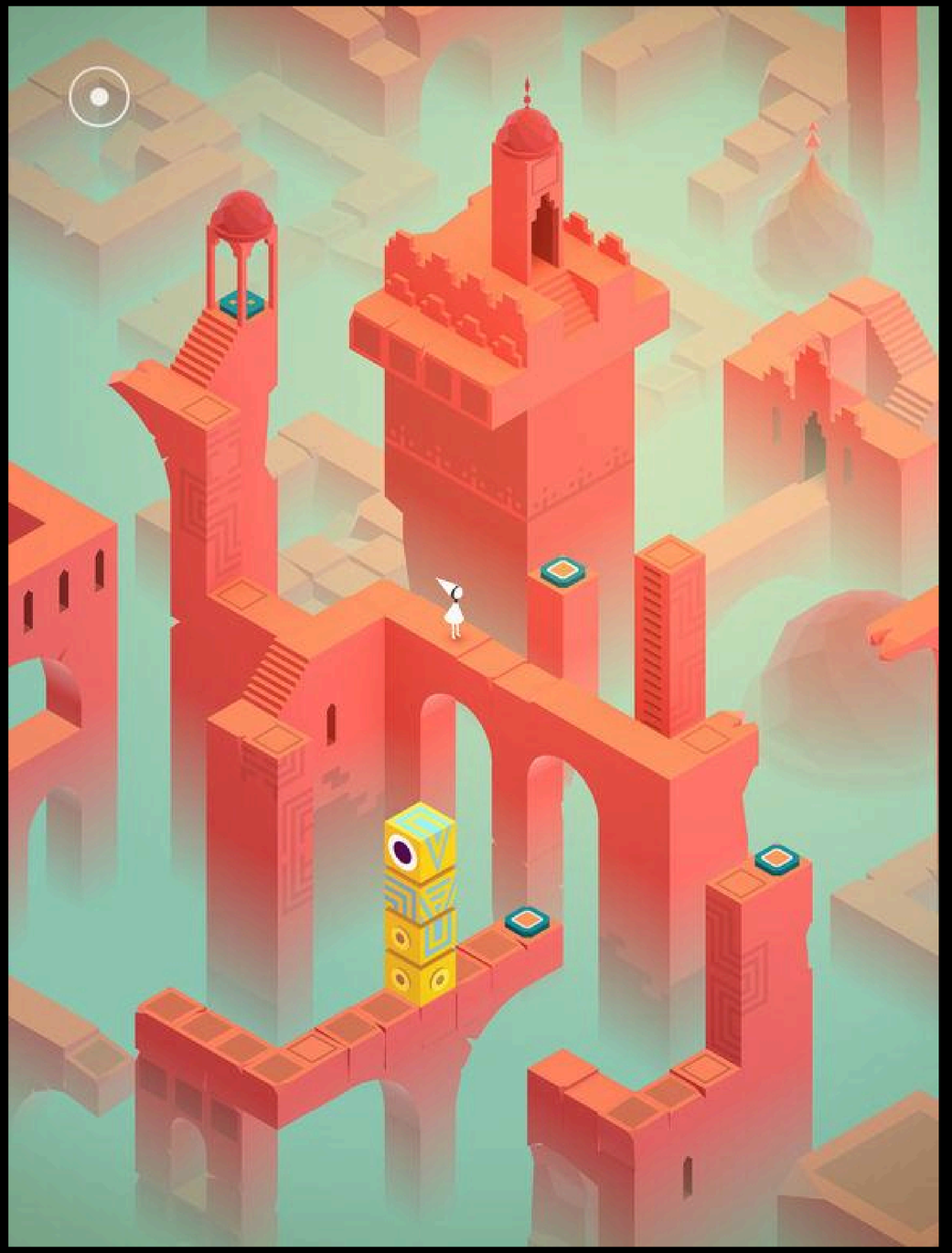
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Shader examples:
Gradients - world position / local position

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Shader examples:
Gradients - world position / local position



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```glsl
#ifdef GL_ES
precision mediump float;
#endif

#define PI 3.14159265359

uniform vec2 u_resolution;
uniform vec2 u_mouse;
uniform float u_time;

float plot(vec2 st, float pct){
  return  smoothstep( pct-0.02, pct, st.y) -
          smoothstep( pct, pct+0.02, st.y);
}

void main() {
    vec2 st = gl_FragCoord.xy/u_resolution;

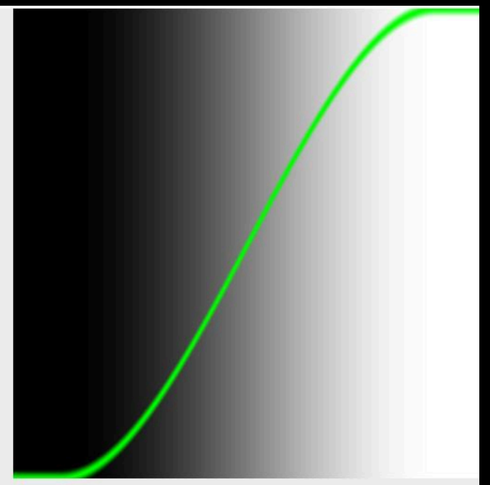
    // Smooth interpolation between 0.1 and 0.9
    float y = smoothstep(0.1,0.9,st.x);

    vec3 color = vec3(y);

    float pct = plot(st,y);
    color = (1.0-pct)*color+pct*vec3(0.0,1.0,0.0);

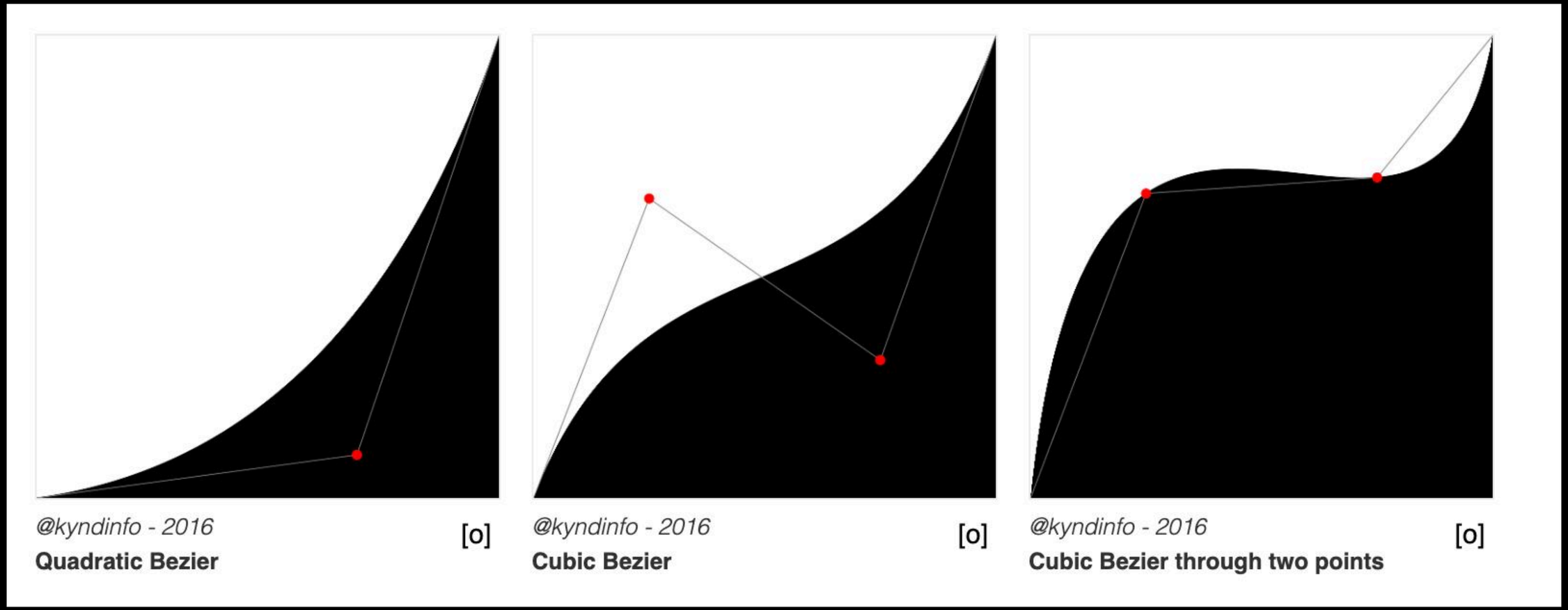
    gl_FragColor = vec4(color,1.0);
}

```



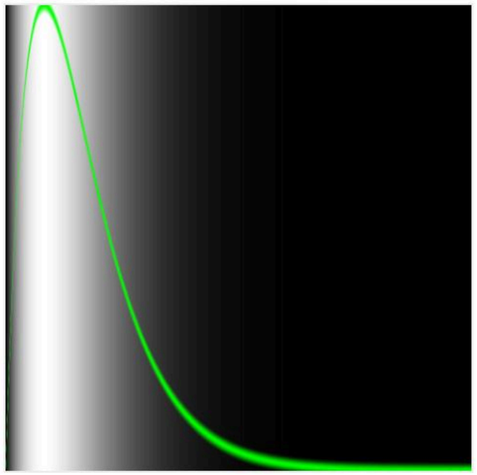
Shader examples:
Gradients - interpolations: linear, smooth

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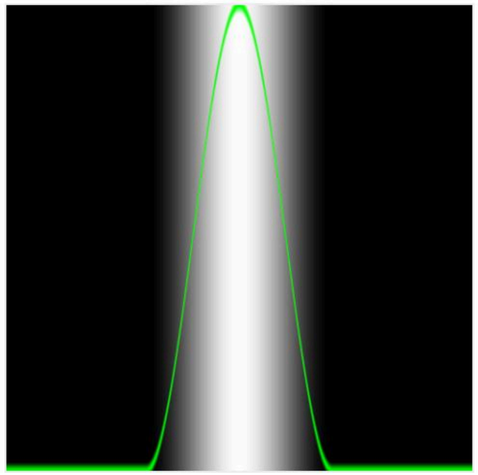


Shader examples:
Gradients - interpolations - bezier

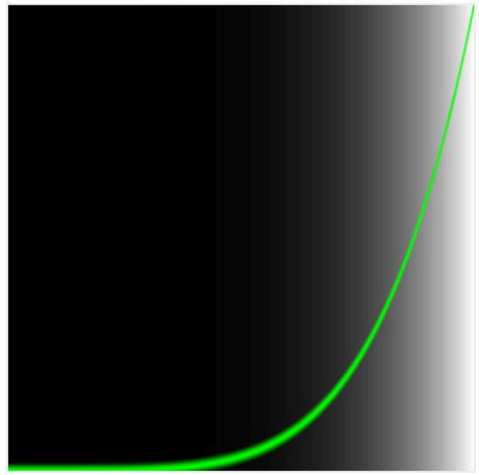
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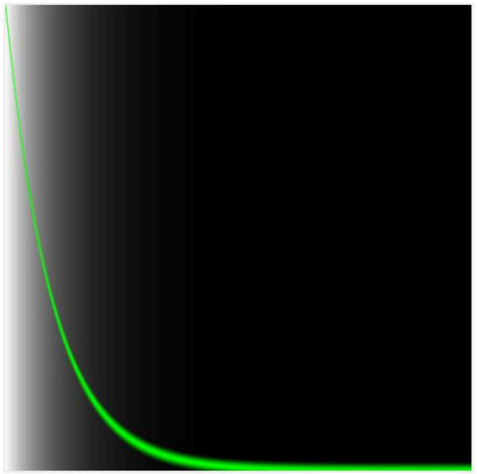
Inigo Quiles [o]
Impulse



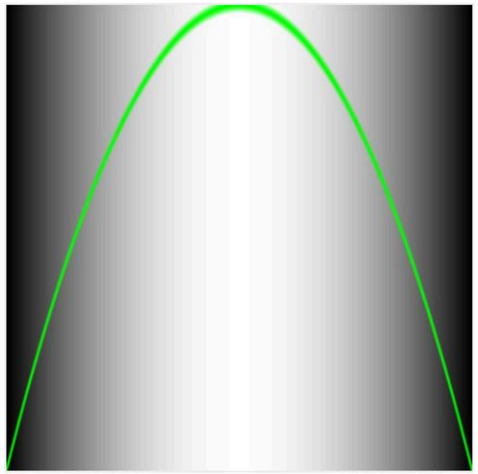
Inigo Quiles [o]
Cubic Pulse



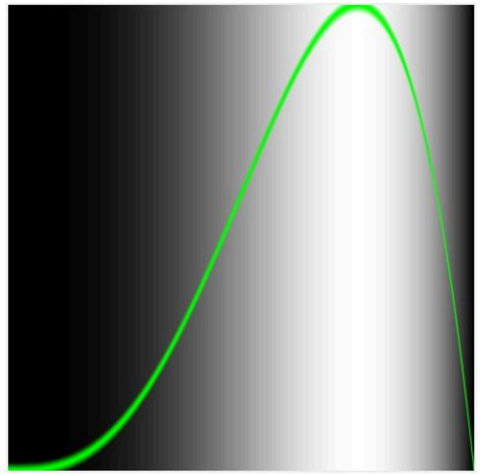
Inigo Quiles [o]
Expo



Inigo Quiles [o]
ExpStep



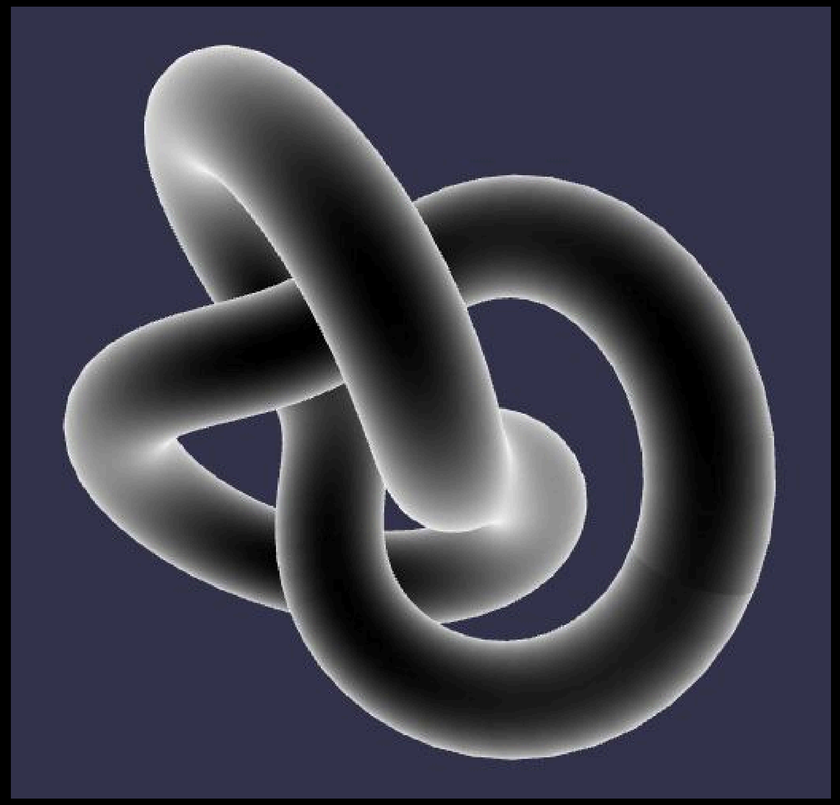
Inigo Quiles [o]
Parabola

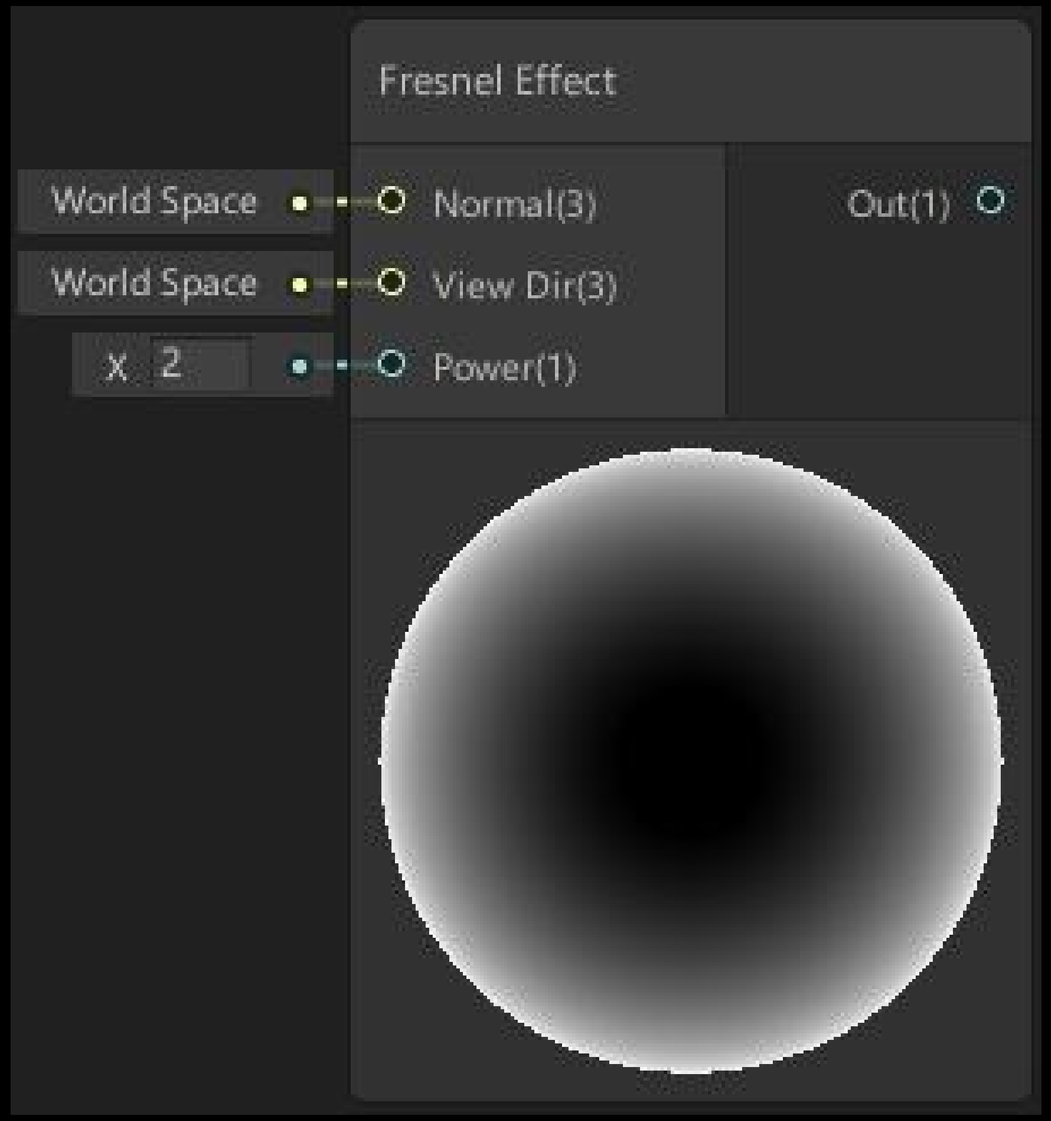


Inigo Quiles [o]
Pcurve

Shader examples:
Gradients - interpolations: equations

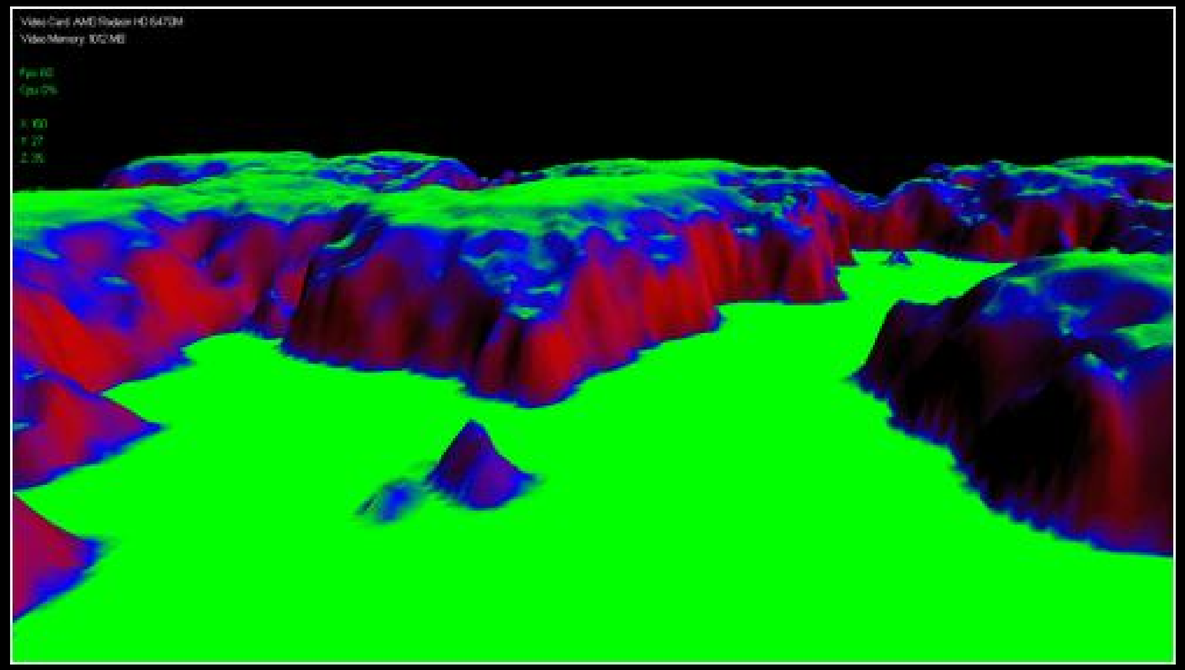
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Shader examples:
Fresnel

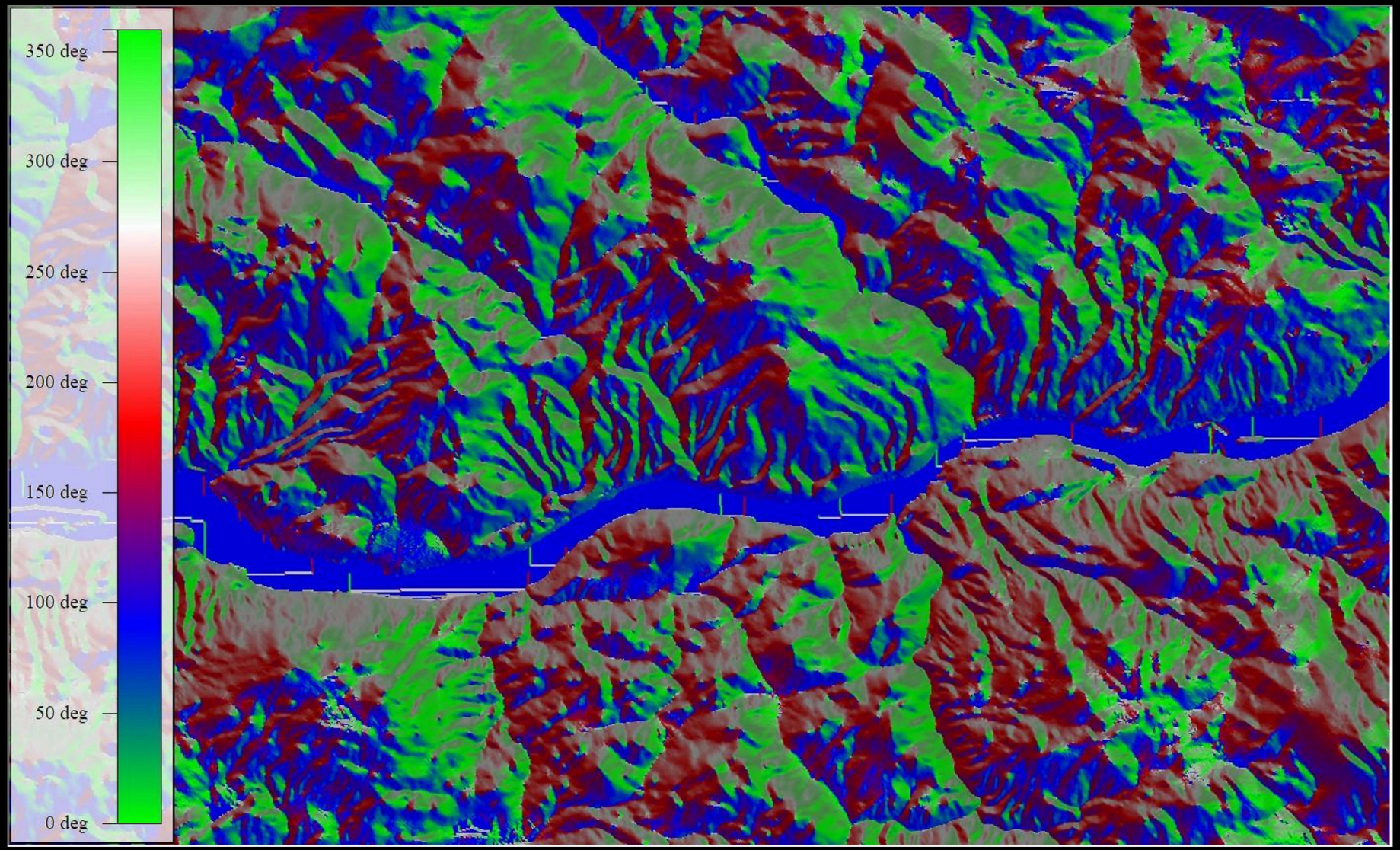
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Shader examples:
Gradients - slope

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Shader examples:
Gradients - slope

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Shader examples:
Distance driven

ref: Islands non places

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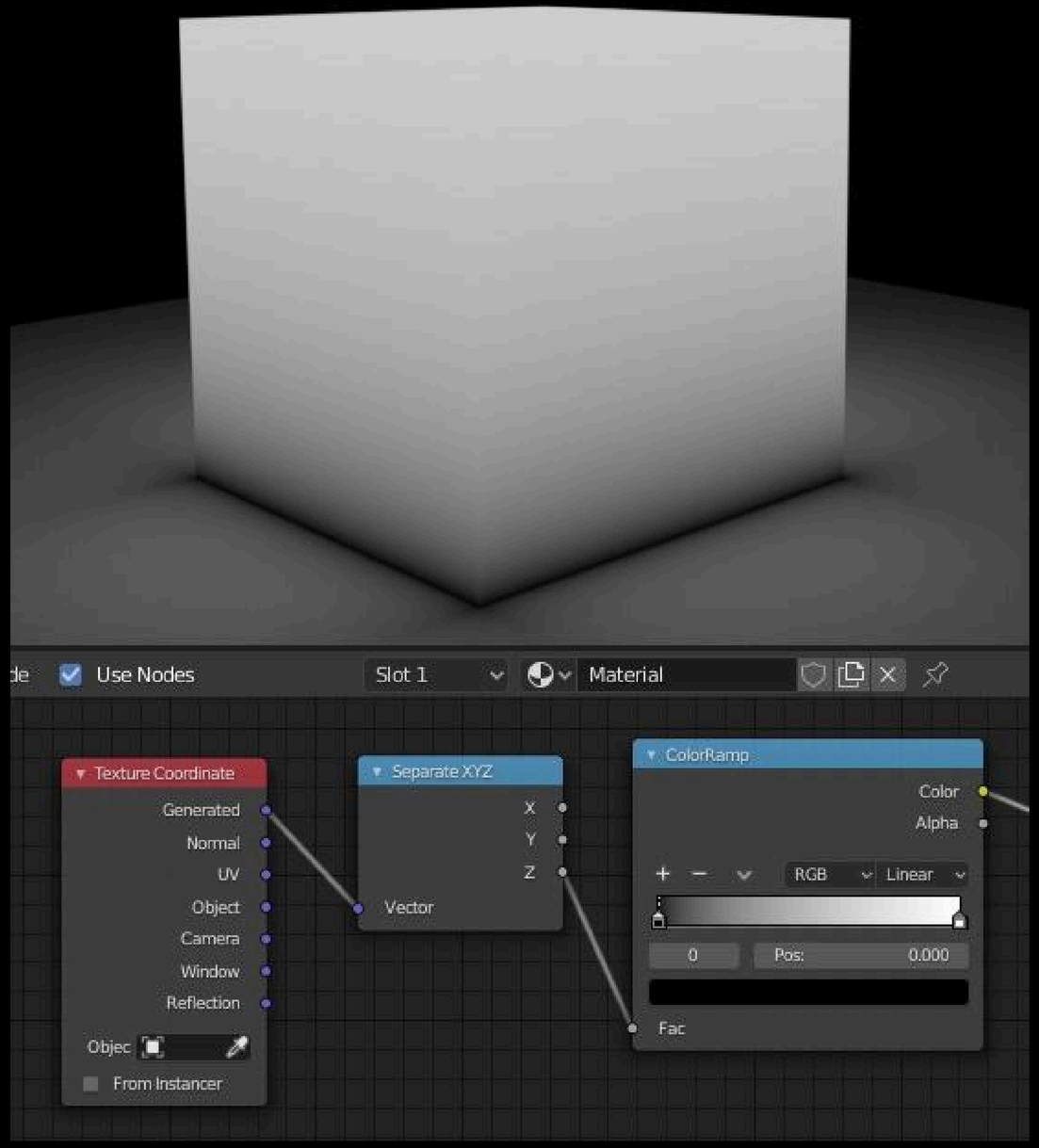


Shader examples:
Distance driven

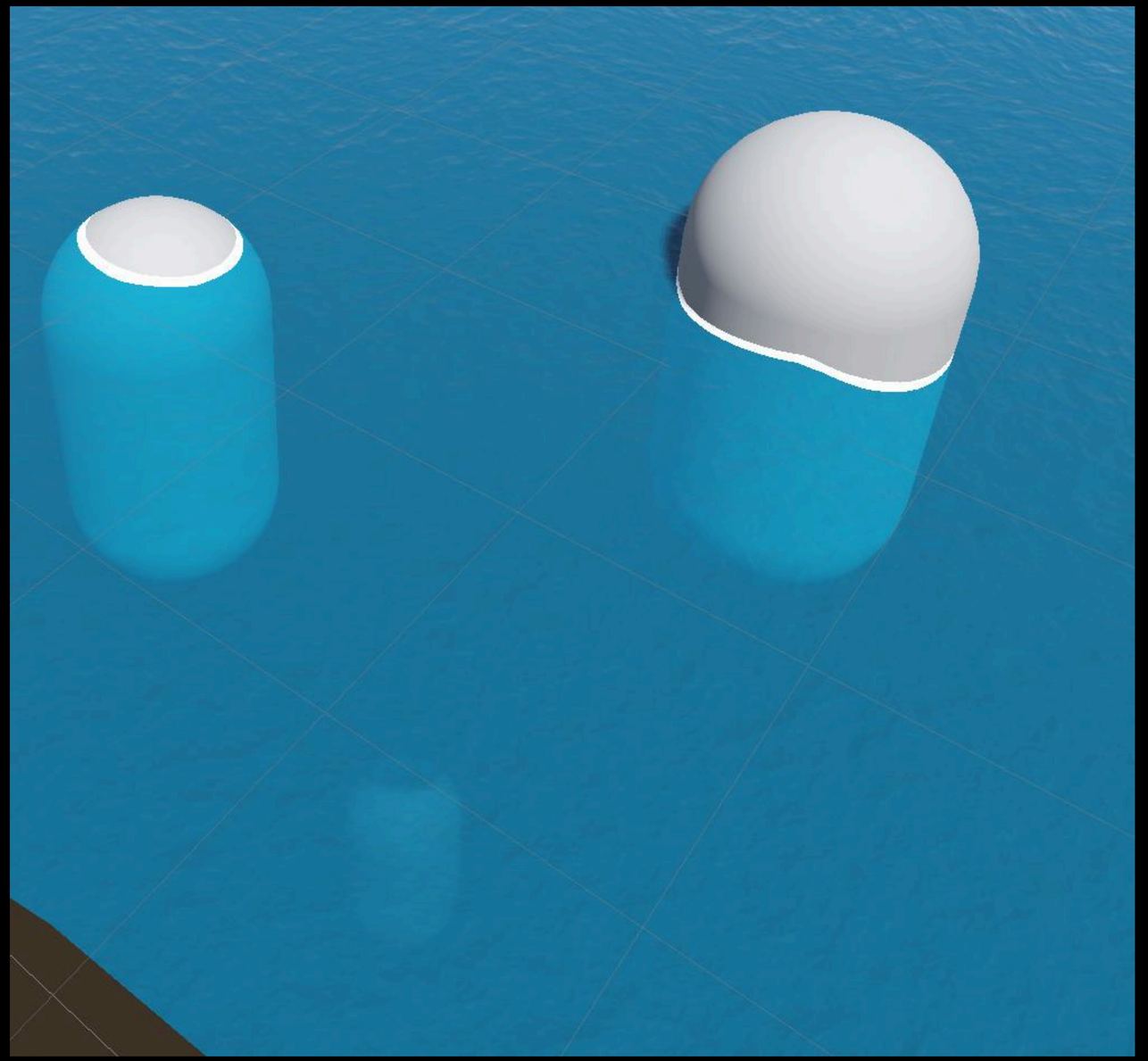
ref: Islands non places

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Shader examples:
Ambient occlusion

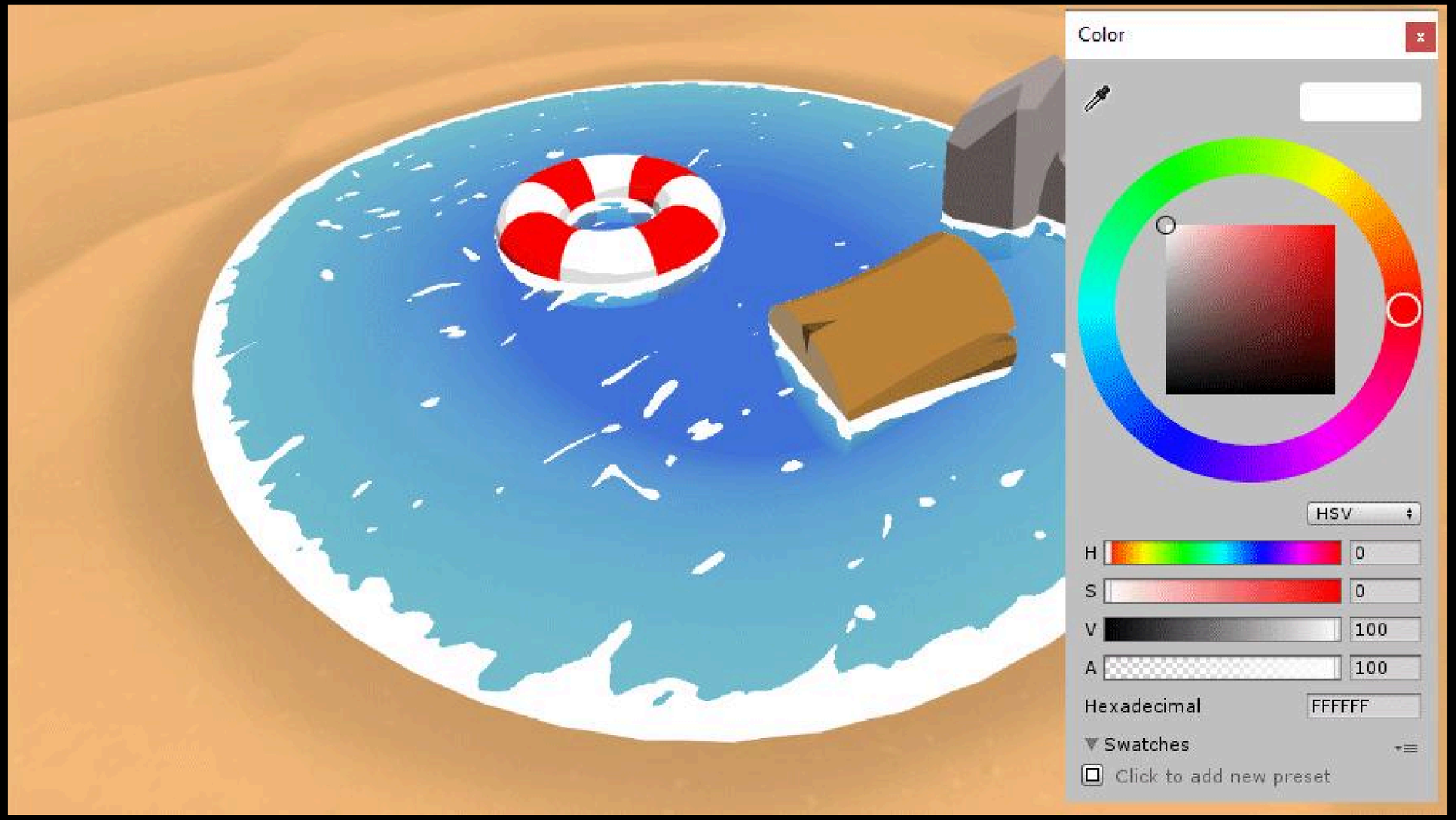


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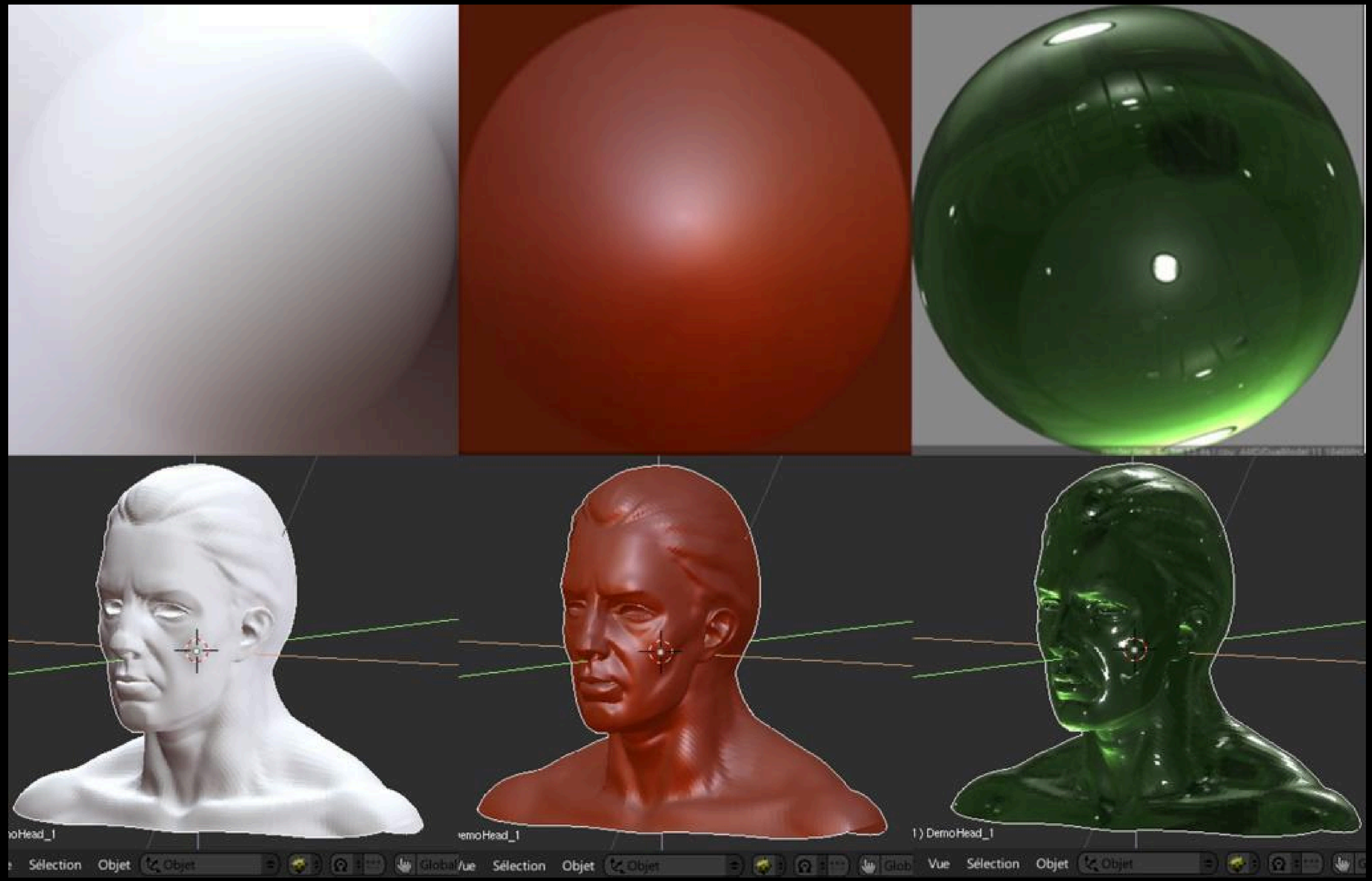
Shader examples:
Contact Shaders

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Shader examples:
Contact Shaders

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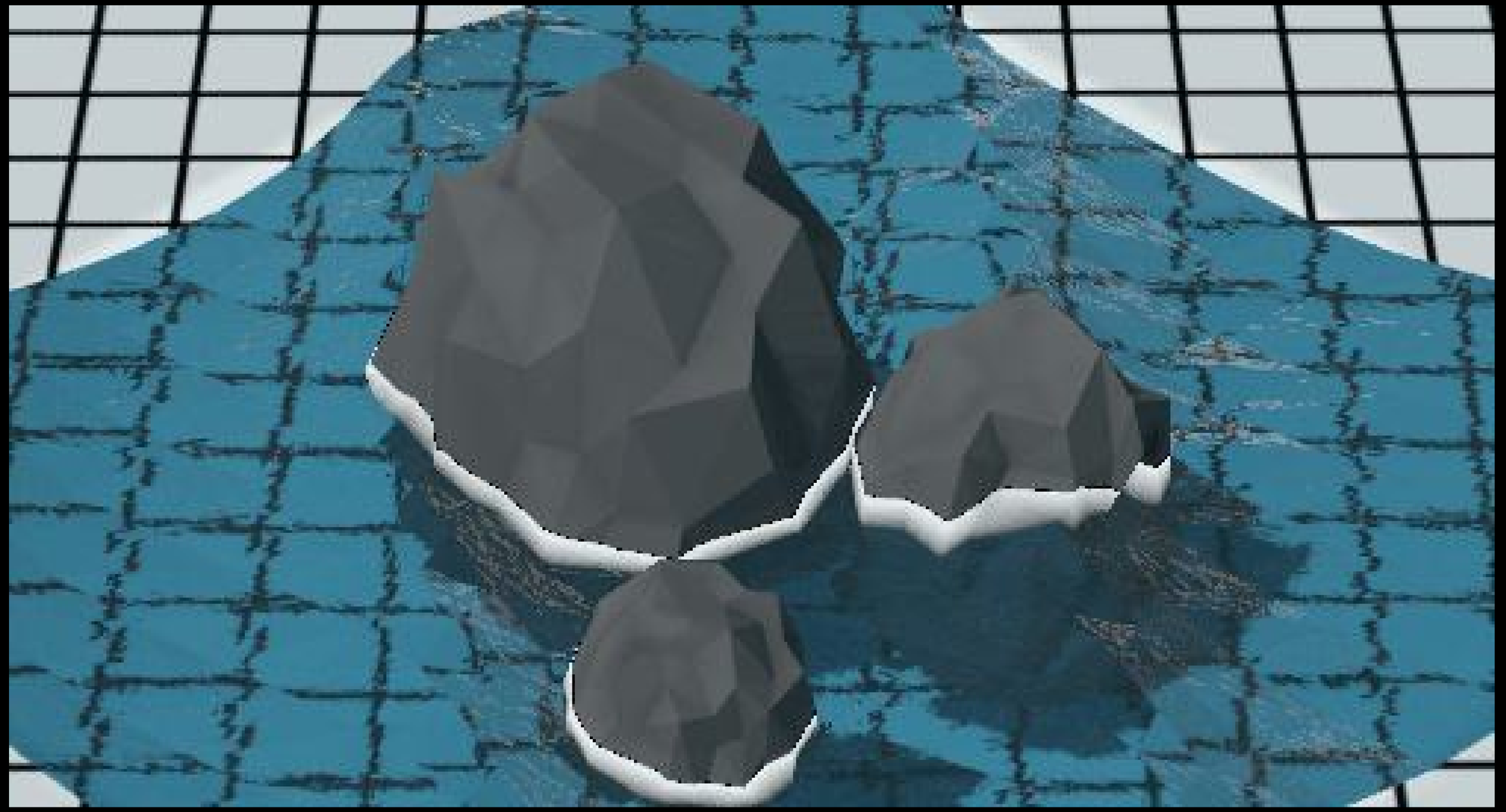
Shader examples: Matcap

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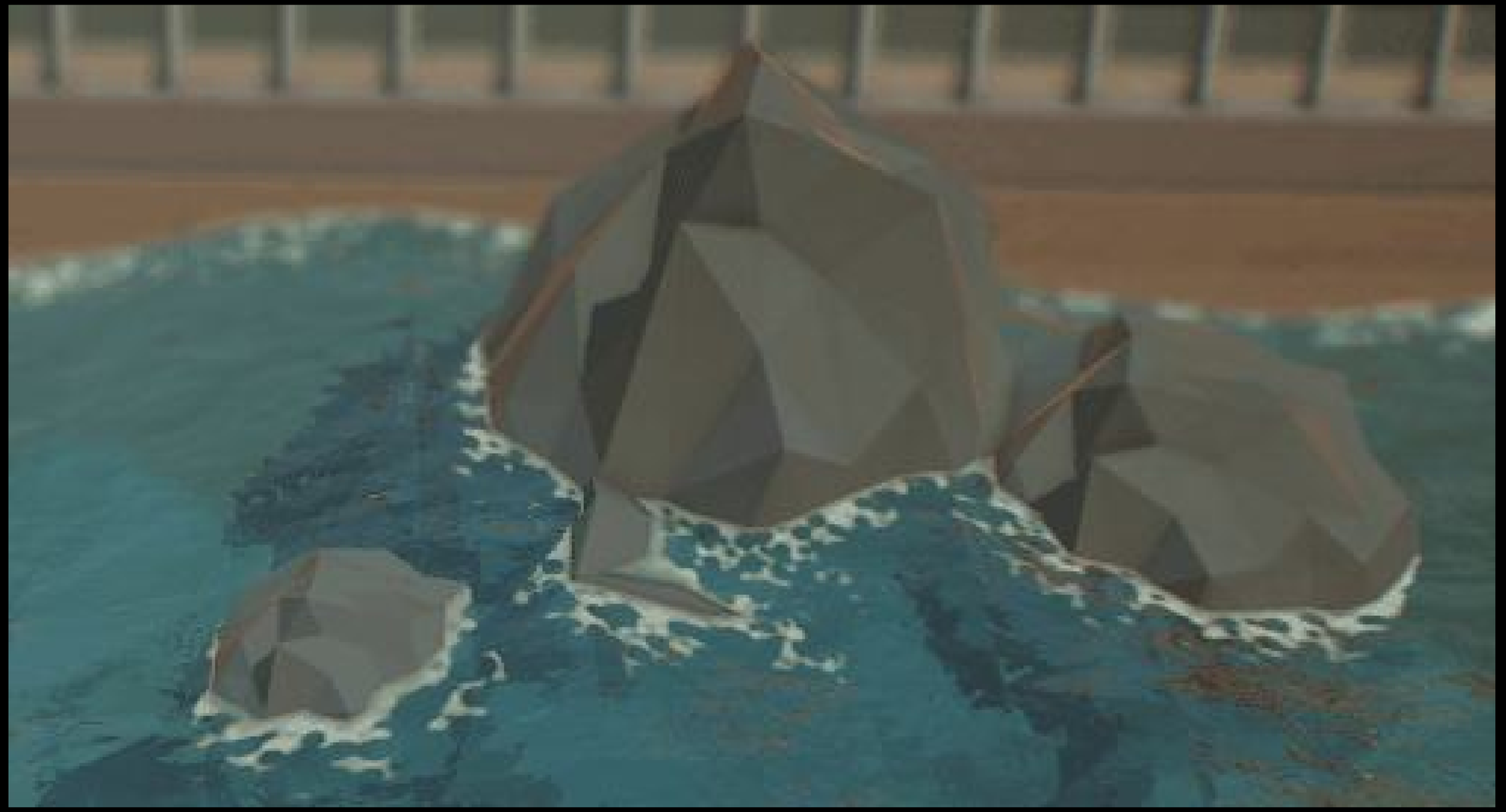
Shader examples:
Vertex displacement - object pop, surface ripple

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Shader examples:
Contact Shaders + vertex deformation

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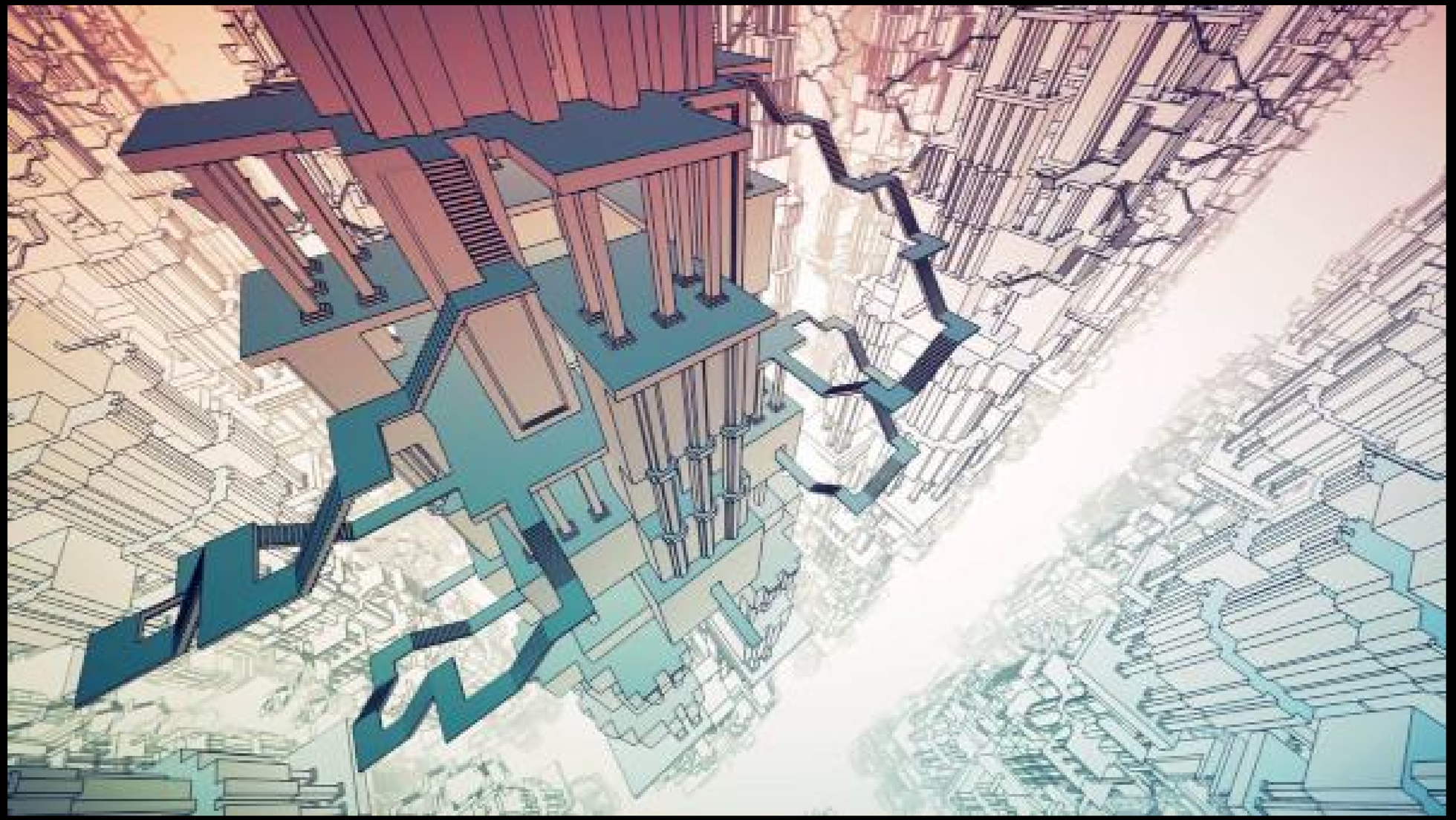
Shader examples:
Contact Shaders + noise + vertex deformation

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Shader examples:
Sable



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Shader examples:
Manifold Garden

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Assignment 2:

-Create your own shader studies
-Demonstrate an understanding of what shaders can do for your simulations
-Consider what shaders you could use or would like to develop, and why
-Implement a section of your app that is dedicated to shaders and that allows you to swap between different strategies.